

Life Expectancy and Sanitation

1. Project Objective

The objective of this project is to predict Life Expectancy in developing nations using:

- Access to Basic Sanitation
- Infant Mortality Rate The study focuses on:
- Improving R^2 model fit
- Reducing prediction errors (MAE, RMSE)
- Understanding the impact of sanitation and mortality on longevity
- Normalizing skewed health indicators

2. Models Implemented

- Simple Linear Regression
- Multiple Linear Regression
- Polynomial Regression

3. Model Performance Results

3.1 Simple Linear Regression

Description

Simple Linear Regression was used to analyze the relationship between:

- Life Expectancy (Target)
- One predictor at a time (e.g., Sanitation or Infant Mortality) **Equation:**

$$Y = \beta_0 + \beta_1 X$$

Purpose in Project

- Understand individual impact of each variable.

- Identify strongest single predictor.
- Provide baseline performance.

Results:

- R² Score: 0.8579
- RMSE: 2.06
- MAE: 1.53

Interpretation:

- The model explains 85.78% of the variation in life expectancy.
- On average, predictions are off by only 1.53 years, which is very accurate.
- RMSE (2.06) being close to MAE indicates fewer large errors.
- This is the best performing model among the three.

Contribution:

- Strong predictive accuracy
- Simple and interpretable
- Demonstrated strong relationship between predictor and life expectancy

3.2. Multiple Linear Regression

Description

MLR was used to analyze combined impact of:

- Sanitation Access
- Infant Mortality **Equation:**

$$LifeExpectancy = \beta_0 + \beta_1(Sanitation) + \beta_2(InfantMortality)$$

Purpose in Project

- Measure combined influence of health indicators.
- Improve prediction accuracy.
- Evaluate overall explanatory power.

Results:

- R² Score: 0.8296
- RMSE: 3.84
- MAE: 2.80

Interpretation:

- Explains 82.96% of variation in life expectancy.
- Average prediction error ≈ 2.8 years.
- Slightly higher RMSE suggests presence of larger prediction errors.

Contribution:

- Captured combined impact of sanitation and infant mortality
- Provided balanced model performance
- Demonstrated that multiple public health indicators influence life expectancy

3.3. Polynomial Regression

Description

Polynomial Regression captures non-linear relationships.

Example (Degree 2):

$$LifeExpectancy = \beta_0 + \beta_1 X + \beta_2 X^2$$

Why Polynomial Was Needed

- Relationship between sanitation and life expectancy may not be strictly linear.
- Health indicators often show diminishing returns.
- Infant mortality effect may be stronger at lower ranges

- **Results (Approximate):**

- R^2 Score: ≈ 0.82
- RMSE: $\approx 3.9\text{--}4.0$
- MAE: $\approx 2.8\text{--}3.0$

- **Interpretation:**
- Explains around 82% of variation.
- Error slightly higher than linear regression.
- Non-linear modeling did not significantly improve performance.

Contribution:

- Tested non-linear relationships
- Confirmed that relationship is largely linear
- Helped validate model structure

4.Comparative Model Analysis

Model	R ²	RMSE	MAE	Performance Level
Simple Linear Regression	0.8579	2.05	1.53	Best
Multiple Linear Regression	0.8045	4.11	2.98	Good
Polynomial Regression	0.8381	3.74	2.21	Moderate

5.Model Evaluation and Performance Analysis

5.1 Simple Linear Regression

```
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("MAE:", mae)
print("RMSE:", rmse)
print("R2 Score:", r2)
```

MAE: 1.5325025530204495
RMSE: 2.0585764084640013
R2 Score: 0.8578825125774813

The model evaluation results show strong performance, with an **MAE of 1.53**, meaning predictions differ from actual life expectancy by about 1.5 years on average, an **RMSE of 2.06**, indicating low overall prediction error, and an **R² score of 0.858**, which means the model explains approximately 85.8% of the variation in life expectancy, demonstrating high predictive accuracy.

5.2 Multiple Linear Regression

```
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("MAE:", mae)
print("RMSE:", rmse)
print("R2 Score:", r2)
```

✓ 0.0s

```
MAE: 2.980629768229549
RMSE: 4.116175806150857
R2 Score: 0.8045065382162837
```

The model evaluation shows moderate performance, with an **MAE of 2.98**, meaning predictions differ from actual life expectancy by about 3 years on average, an **RMSE of 4.12**, indicating the presence of some larger prediction errors, and an **R² score of 0.8045**, which means the model explains approximately 80.45% of the variation in life expectancy, demonstrating good but lower accuracy compared to the previous model.

5.3. Polynomial Regression

```
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error
import numpy as np

poly = PolynomialFeatures(degree=2)

x_train_poly = poly.fit_transform(x_train)
x_test_poly = poly.transform(x_test)

model = LinearRegression()
model.fit(x_train_poly, y_train)

# Predictions
y_pred = model.predict(x_test_poly)

r2 = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))

print("R2 Score :", r2)
print("MAE      :", mae)
print("RMSE     :", rmse)
```

✓ 0.1s

```
R2 Score : 0.8381308901031268
MAE      : 2.214753442949079
RMSE     : 3.7454998703807107
```

The Polynomial Regression model (degree 2) achieved an **R² score of 0.8381**, meaning it explains about 83.81% of the variation in life expectancy, with an **MAE of 2.21** and an **RMSE of 3.75**, indicating moderate prediction error; overall, the model performs well but does not outperform the best linear regression model

6 Key Findings

- Simple Linear Regression achieved the highest R² and lowest error.
- Multiple Linear Regression slightly reduced accuracy compared to simple linear.
- Polynomial Regression did not improve performance significantly.
- The relationship between health indicators and life expectancy appears mostly linear.
- Infant mortality and sanitation remain strong determinants of longevity.

7 Which Model Best Achieved the Objective?

Best Model: Simple Linear Regression

- Highest R² (85.78%)
- Lowest MAE (1.53 years)
- Lowest RMSE (2.06)

8 Alignment with SDG 3

- Improvements in sanitation and reduction in infant mortality significantly increase life expectancy.
- Public health indicators strongly influence national longevity.
- Data-driven modeling can support health policy decisions.

9 Conclusion

The study successfully modeled life expectancy using regression techniques.

- Simple Linear Regression provided the most accurate predictions.
- Multiple Linear Regression confirmed the combined impact of health indicators.
- Polynomial Regression showed that non-linear effects were minimal.
- Life expectancy in developing nations is strongly influenced by key public health variables, and linear modeling effectively captures this relationship.

10 Learning Outcomes

- Applied Linear, Multiple, and Polynomial Regression
- Interpreted R^2 , MAE, and RMSE
- Understood model comparison and validation
- Learned impact of skewness and normalization
- Connected data science analysis with SDG policy goals